Infant Contingency Learning

Infants can learn that a self-generated action predicts a sensory consequence in the mobile-conjugate-reinforcement paradigm; this is action–outcome learning under controlled conditions, not adult-like causal theory or a stimulation mandate.

GRADE: HIGH-CONTROLLED; LOW-TRANSFER | STATUS: STABLE | DOMAIN: LEARNING, PLAY | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

In the mobile-conjugate-reinforcement paradigm, a ribbon links an infant's ankle to an overhead mobile. Kicking moves the mobile; later kicking above baseline indexes retention of that action–outcome relation. The design manipulates whether the infant's own action produces the consequence, which is stronger identification than passive observation.

EVIDENCE AND CAUSAL ASSESSMENT

Rovee & Rovee (1969) established conjugate reinforcement of infant kicking. Subsequent work, summarized in methods papers (e.g. Merz et al. 2017) and a 2025 review of the last quarter-century, treats the paradigm as a reliable laboratory measure of learning and memory: speed of acquisition and length of retention increase with age; reminders can reactivate a forgotten response, often with high context specificity. Incomplete processing parameters and the later turn toward individual differences and agency do not block the narrow High claim that the contingency can be acquired and retained under matching conditions — the same split as Saffran laboratory segmentation versus household transfer.

EFFECT SIZE AND PERSISTENCE

There is no single standardized effect for "infant learning." The defensible unit is behavioral acquisition and a retention window that varies with age, training distribution, and contextual match. A weak test response can be retrieval or motor-performance failure rather than absent learning.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (laboratory operant conditioning / infant contingency reinforcement, not an observational population phenotype association).

Arousal, interest in the mobile, motor capacity, and the opportunity to kick all affect the score. Laboratory contingency is not everyday exploration. Agency (whether the infant experiences the relation as self-caused) remains an interpretive question.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean infants are miniature scientists, that every repeated action is conscious goal pursuit, or that parents must run training protocols. Early task performance is not destiny.

WHAT ACTION IS REASONABLY JUSTIFIED

Ordinary opportunities to act and notice consequences are consistent with the finding. Intensive stimulation programs and performance monitoring do not follow.

VERIFIED SOURCES & DOI IDENTIFIERS

- Rovee & Rovee (1969), Journal of Experimental Child Psychology
- Merz et al. (2017), mobile-paradigm methods (PMC5716629)
- Cuevas et al. (2025), Infant Behavior & Development (PMC12354038)